

Outlier Detection & EDA

Week 3 - Day 4

The Odd One Out:

Understanding Outliers

Imagine you're analyzing the heights of a basketball team. Most players are between 6'5" and 7'0", but one player stands at 5'4". This player is an outlier—a data point that significantly deviates from the norm. Outliers can arise due to:

Measurement errors: Faulty equipment or data entry mistakes.

Natural variations: Genuine but rare occurrences in data.



What You'll Learn



Outlier Detection



Why Outlier Detection is critical for ML success?

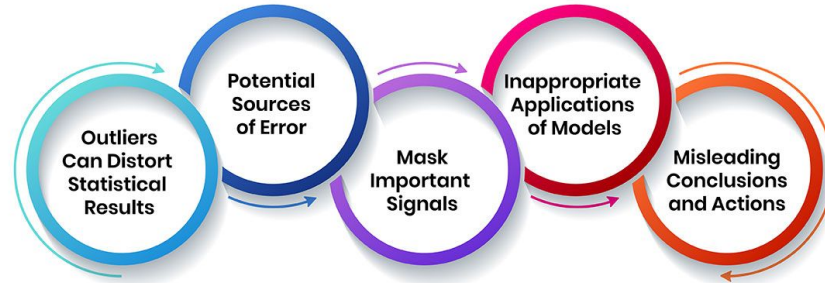


Why Do Outliers Matter?

Outliers can profoundly impact machine learning models:

- **Skewed Analysis:** They can distort statistical measures like mean and standard deviation.
- **Model Performance:** Algorithms may fit to outliers, reducing generalization to new data.
- **Misleading Insights:** Outliers can lead to incorrect conclusions if not addressed.

The Importance of Outlier Detection in Data Analysis



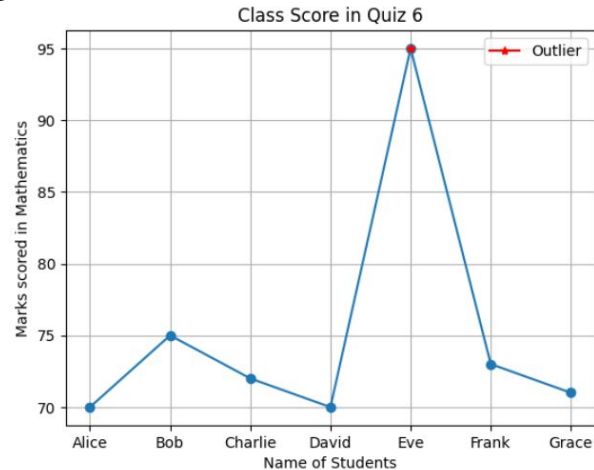
Types of Outliers: Global vs. Contextual

1

Global Outliers:

Data points that are extreme across the entire dataset.

Example: A temperature reading of 100°C in a dataset where the average is 22°C.

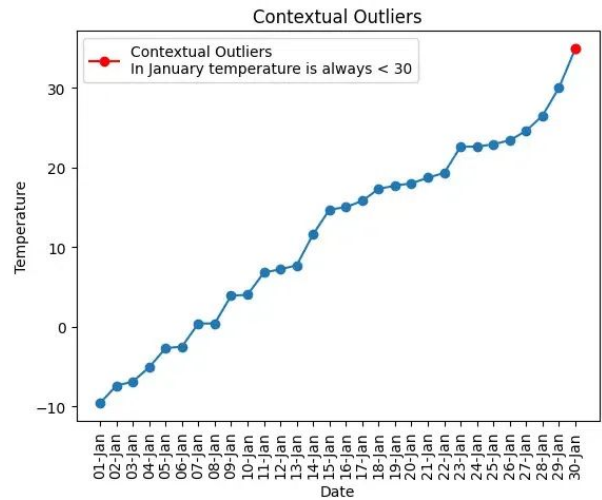


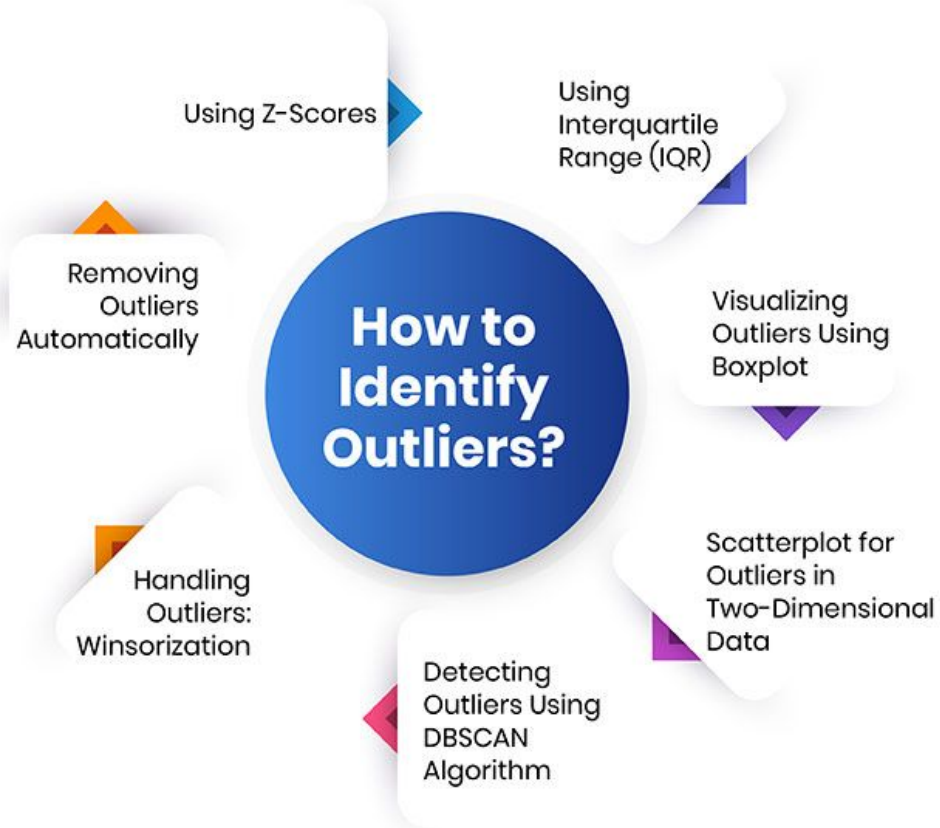
2

Contextual Outliers:

Data points that are unusual in a specific context but may be normal otherwise.

Example: High spending on luxury items during holiday seasons.





Detecting Outliers: The Z-Score Method

The Z-score measures how many standard deviations a data point is from the mean. It's calculated as:

$$Z = \frac{(X - \mu)}{\sigma}$$

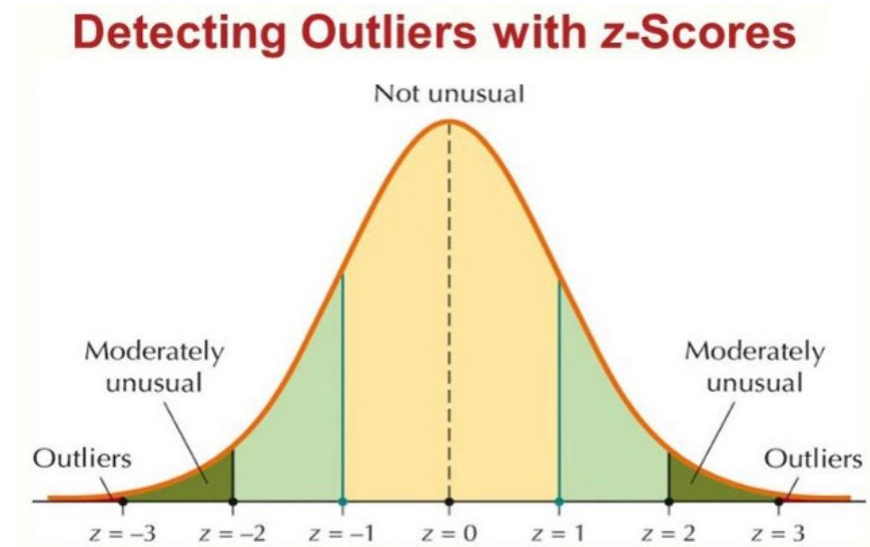
Where:

X = data point

μ = mean of the dataset

σ = standard deviation

A common threshold is $|Z| > 3$, indicating a potential outlier.



Applying Z-Score in Python

```
import numpy as np
from scipy.stats import zscore

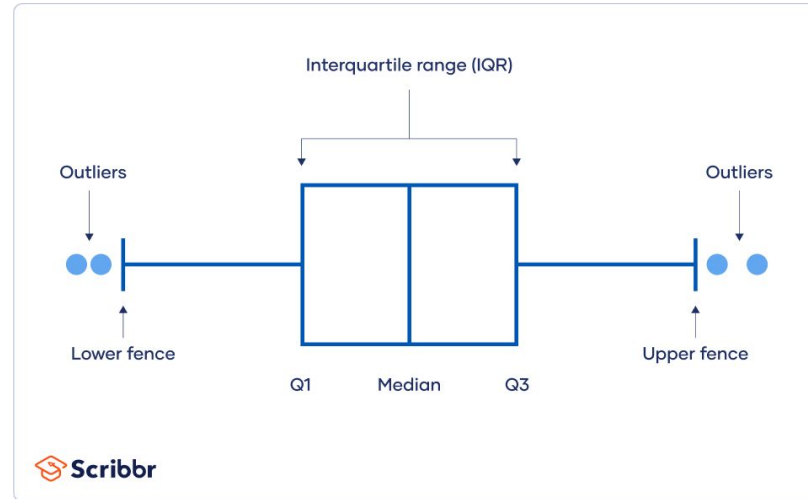
data = np.array([10, 12, 12, 13, 12, 11, 112])
z_scores = zscore(data)
outliers = np.where(np.abs(z_scores) > 3)
print(outliers)
```

This code identifies indices of data points where the absolute Z-score exceeds 3.

Detecting Outliers: The Interquartile Range (IQR) Method

The **IQR method** identifies outliers by focusing on the middle 50% of the data:

1. **Calculate Q1 (25th percentile) and Q3 (75th percentile):**
 - $IQR = Q3 - Q1$
2. **Determine the bounds:**
 - Lower Bound: $Q1 - 1.5 \times IQR$
 - Upper Bound: $Q3 + 1.5 \times IQR$
3. **Identify data points outside these bounds as outliers.**



Applying IQR in Python

```
import numpy as np

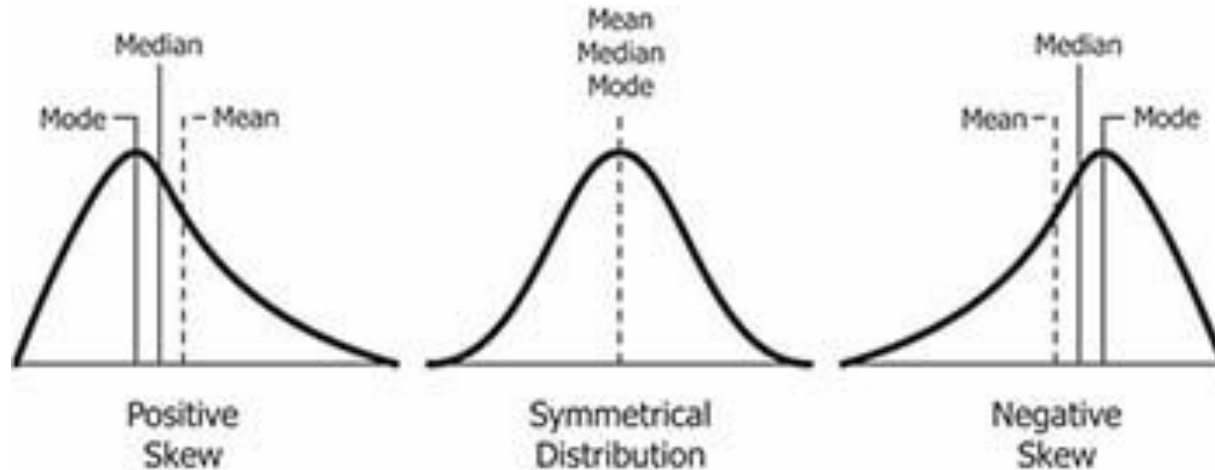
data = np.array([10, 12, 12, 13, 12, 11, 112])
Q1 = np.percentile(data, 25)
Q3 = np.percentile(data, 75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
outliers = np.where((data < lower_bound) | (data > upper_bound))
print(outliers)
```

This script identifies indices of data points outside the calculated bounds.

Z-Score vs. IQR – Which One?

Z-Score: Best for **normally distributed** data

IQR: Works well for **skewed or non-normal** distributions

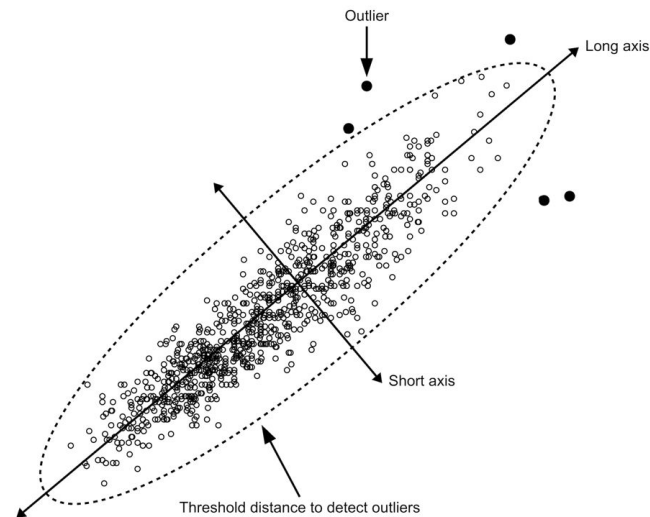
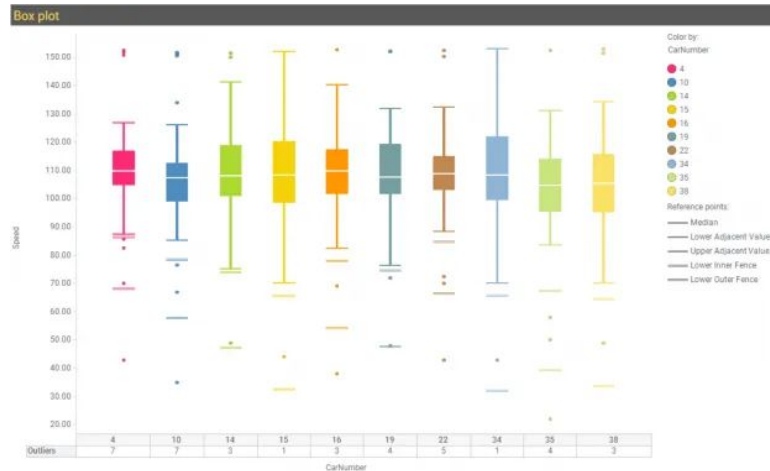


Visual Detection: Spot the Outlier

Sometimes, a good chart tells you more than a formula.

Use:

- **Boxplots** – great for 1D numeric variables
- **Scatter plots** – see relationships and spot weird patterns



What To Do With Outliers?

Outliers aren't always enemies. You have options:

1. **Keep** – If meaningful (e.g., luxury home price)
2. **Remove** – If due to error or irrelevant
 - Pros:** Simplifies data, removes noise.
 - Cons:** Risk of losing valuable information.
3. **Transform** – Use log or robust methods to reduce impact

Scaling: Normalize data to reduce outlier impact.

Winsorization: Limit extreme values to reduce effect without removal.

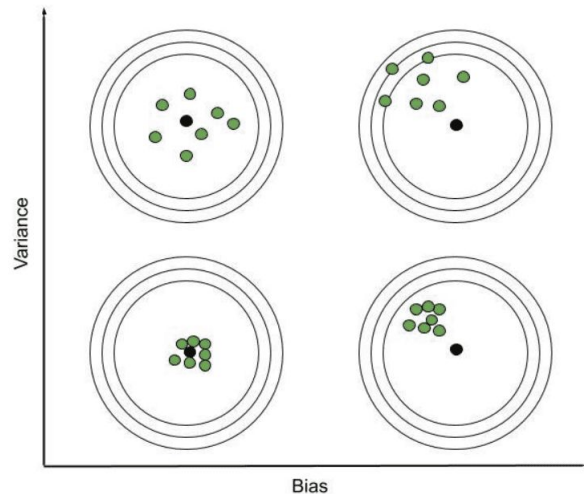
Log Transformation: Compresses range, mitigating impact of large outliers.



Bias and Variance: The Two Faces of Model Error

Imagine you're shooting arrows at a target 🎯. Your model's **accuracy** is how close those arrows are to the center.

- **Bias**: All arrows consistently miss the target in the same direction → the model is **too simple** and underfits the data.
- **Variance**: The arrows are all over the place, even if some hit the center → the model is **too complex** and overfits.



What Does This Have to Do With Outliers?

Outliers can **increase model variance**. If your model tries to fit these rare, extreme points:

- It becomes overly sensitive to noise.
- Small changes in the data → big changes in predictions.
- You lose generalization on real-world data.

On the other hand, ignoring important outliers can **introduce bias**, especially if those outliers represent valid edge cases (like fraudulent transactions or rare diseases).

